

RISLAN MOHAMED

Software Engineer | AI/ML Developer | Cybersecurity Researcher | Artificial Intelligence & Data Science | Data Analytics | Business Intelligence Developer

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PROFESSIONAL SUMMARY

Results-driven Associate Software Engineer with hands-on experience in full-stack web development, AI/ML systems, database management, and cybersecurity. Skilled in building intelligent applications using TensorFlow, Claude AI, and Python, with a strong foundation in penetration testing and network infrastructure. Completed my Pearson BTEC Higher National Diploma in Computing (Software Engineering), combining academic research with real-world industry practice. Passionate about leveraging artificial intelligence to solve complex security and operational challenges, currently pursuing my BSc (Hons) Computer Science in Artificial Intelligence and Data Science (Top-Up Degree).

TECHNICAL SKILLS

AI / Machine Learning: *TensorFlow, Deep Learning, Scikit-Learn, Predictive Modeling, Model Evaluation*

Data Analytics & BI: *Data Cleaning, Data Analysis, Interactive Dashboards, Demand Forecasting, Power BI*

Data Visualization: *Matplotlib, Seaborn, Plotly, Streamlit, Chart.js, Jupyter Notebook*

Programming Languages: *Python, Java, JavaScript, SQL, HTML, PHP, CSS*

Databases: *MySQL, phpMyAdmin, H2 Database, Relational Database Design & Optimization*

Data Engineering Tools: *Pandas, NumPy, REST APIs, Spring Boot, Selenium, JUnit 5, PyTest*

Cybersecurity (Supporting): *Penetration Testing, Kali Linux, CVE Analysis, Network Security*

Other Tools: *Git, GitHub, Claude AI, Chat GPT, Model Context Protocol (MCP), Canva*

PROFESSIONAL EXPERIENCE

Associate Software Engineer

November 2025 - present

UNIX-NET-TECHNOLOGIES

- Build responsive web interfaces using HTML5, CSS3, Bootstrap, and JavaScript.
- Develop secure PHP backend solutions and manage MySQL databases.
- Handle web hosting, domain management, DNS configuration, and performance optimization.

Website Manager

May 2025 – Present (Seasonal)

EuroFresh Finland (Online)

- Manage e-commerce platform — product listings, pricing, and inventory updates.
- Maintain database accuracy to ensure reliable product data for customers.

Freelance Software Engineer

March 2025 – 2026

Independent Consultant

- Deliver custom software, SQL databases, and network infrastructure for clients.
- Design VLANs, subnetting, and security configurations; produce technical documentation.

IT Administrator & Analyzer

2023 – 2025

Afran Travels & Tours

- Maintained systems and network infrastructure ensuring 99%+ uptime.
- Enforced cybersecurity policies, managed user accounts, and provided technical support.

RESEARCH & PROJECTS

AI-Powered Automated Penetration Testing System

Nov 2025 – Jun 2026

Technologies: Python · Claude AI · MCP · Kali Linux · CVE RAG · DeepSeek-V3.2

- Built an AI-assisted penetration testing pipeline integrating Claude AI, MCP, and Kali Linux to automate vulnerability scanning, enrichment, and analysis.
- Developed a local CVE RAG engine enriching scan outputs with CVSS scores, severity ratings, and remediation recommendations.
- Evaluated using Precision, Recall, F1 Score, and Confusion Matrix; validated with DeepSeek-V3.2. (Final Thesis – BTEC HND, BCAS Campus)

TransitFlow Analytics System

Apr 2026 – Jun 2026

Technologies: Java 21 · Spring Boot · Python · Streamlit · H2 Database · REST API · Pandas · Plotly · JUnit 5 · PyTest · Selenium · MVC · SOLID Principles

- Developed a business intelligence platform capable of processing large-scale transportation and e-commerce datasets using memory-efficient techniques.
- Implemented analytics modules for revenue tracking, customer segmentation, demand forecasting, and logistics performance evaluation through interactive dashboards.
- Applied object-oriented design principles, design patterns, and automated testing frameworks to enhance system scalability, maintainability, and reliability.

AI-Powered Fabric Type Identifier for E-Commerce

April – July 2025

Technologies: Python · TensorFlow · Streamlit · Scikit-Learn · NumPy · Pandas · Matplotlib

- Developed a TensorFlow deep learning model to detect cotton fabric type and estimate cotton percentage with 95.7% accuracy.
- Integrated into a Streamlit app with real-time verification, confidence visualizations, downloadable reports, and watermarked outputs.
- Reduced customer return rates by providing AI-backed product verification on apparel e-commerce platform.

EDUCATION

BSc (Hons) Computer Science in Artificial Intelligence and Data Science (Top-Up Degree)

August 2026 – Present

BCAS Campus, affiliated with Goldsmiths, University of London

- Programmes focuses on machine learning, predictive analytics, big data technologies, data mining, and artificial intelligence, developing intelligent systems that support business decision-making and innovation.

Pearson BTEC Higher National Diploma – Computing (Software Engineering)

Nov 2024 – July 2026

BCAS Campus (Pearson College London)

- Level 5 qualification covering software development, database systems, AI/ML, and cybersecurity. Served as Batch Representative and active member of BCAS IT Society (BITS).

GCE Advanced Level – Bio Science Stream

2021 – 2023

Addalaichchenai MMV National School

- Completed A/Ls in Biology, Chemistry, and Physics; developed strong analytical and scientific reasoning skills.

GCE Ordinary Level

2018 – 2020

Addalaichchenai MMV National School

- Completed O/Ls developed strong analytical and scientific reasoning skills.

CERTIFICATIONS

- Anthropic – Claude Code in Action (February 2026) (Credential ID: 45g2nmewtct9)
- UniAthena – Diploma in Data Visualization Using Python (Score: 89%)
- ADBI E-Learning – Cybersecurity: Principles and Data Security Tools
- Simplilearn – Introduction to AI · Cyber Security 101 · Digital Security Fundamentals · UI/UX Basics
- Additional – Cloud Computing · Machine Learning · Web Development

VOLUNTEERING & LEADERSHIP

- Batch Representative – BCAS IT Society (BITS), 2025–2026: Coordinated between students and faculty; helped organise tech events, workshops, and knowledge-sharing sessions.